

Question-1

	e_1	e_2	e_3	e_4	e_5	e_6
a	1	-1	0	1	-1	0
b	-1	1	-1	0	0	0
c	0	0	0	0	1	1
d	0	0	1	-1	0	-1

a). Matrix M represents a directed graph

- There are -1 in matrix M, which represents edge e_j leaves the vertex v_i .
- There are +1 in matrix M, which represents edge e_j enters the vertex v_i .
- 0, means vertex isn't incident to edge.

b). Adjacency matrix

As per column 1

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \begin{matrix} e_1 \\ -1 \\ 0 \\ 0 \end{matrix}$$

$$b \xrightarrow{e_1} a$$

As per column 2

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \begin{matrix} e_2 \\ -1 \\ 1 \\ 0 \end{matrix}$$

$$a \xrightarrow{e_2} b$$

As per column 3

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \begin{matrix} e_3 \\ 0 \\ -1 \\ 1 \end{matrix}$$

$$b \xrightarrow{e_3} d$$

As per column 4

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \begin{matrix} e_4 \\ 1 \\ 0 \\ 0 \end{matrix}$$

$$d \xrightarrow{e_4} a$$

As per column 5

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \begin{matrix} e_5 \\ -1 \\ 0 \\ 1 \end{matrix}$$

$$a \xrightarrow{e_5} c$$

As per column 6

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \begin{matrix} e_6 \\ 0 \\ 0 \\ -1 \end{matrix}$$

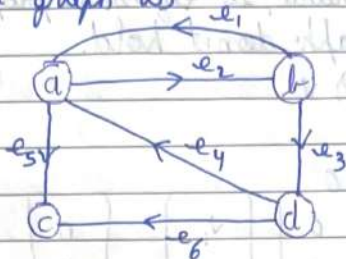
$$d \xrightarrow{e_6} c$$

So, from above details, will construct adjacency matrix

$$\begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

c). Draw graph

As per, week-5, "Matrix representation of directed graphs", will plot graph as



$m_{ij} \rightarrow -1$ if edge e_j leaves vertex v_i ($\rightarrow$)

$m_{ij} \rightarrow 1$ if edge e_j enters vertex v_i ($\leftarrow$)

d). To calculate paths of length 2 are there b/w nodes b & c

$$A^2 = A \cdot A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

a b c d

There are 2 paths b/w b & c of length 2, i.e.

$b \rightarrow a \rightarrow c$, $b \rightarrow d \rightarrow c$

e). $\text{tr}(A^2) = 1 + 1 + 0 + 0 = 2$

$\text{tr}(A^2)$ represents trace of matrix A^2 which is sum of diagonal elements. In terms of connectivity of graph $\text{tr}(A)$ indicates num. of closed walks of length 2 in graph. It provides info. about existence of cycles of length 2 in graph.

f). As, $\text{tr}(A) = 0 + 0 + 0 + 0 = 0 \Rightarrow$ (sum of diagonal elements)

(i) This means sum of eigenvalues is 0. Since, A is real matrix, eigenvalues can be real or complex

(ii) Eigenvalues of A could be deduced if 1-eigenvalue is obvious.

It is likely that 0 is one of the eigenvalues due to trace
 if $\lambda = 0$ is eigenvalue, we solve $Av = 0$
 Given A , isn't symmetric, so eigenvalue properties of
~~undirected~~ undirected graphs don't hold.

$$Av = \lambda v$$

$$\text{if } \lambda = 0$$

$$Av = 0$$

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Equation derived from above

$$v_2 + v_3 = 0 \Rightarrow v_2 = -v_3$$

$$v_1 + v_4 = 0 \Rightarrow v_1 = -v_4$$

$$v_1 + v_3 = 0 \Rightarrow v_1 = -v_3$$

$$\text{So, } v_3 = v_4 = 1 \quad \& \quad v_1 = v_2 = -1$$

$$v = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

Question-2

Let's define the events

- Having social media account $\rightarrow S$
- Not having social media account $\rightarrow S'$
- Having obesity $\rightarrow O$
- Not having obesity $\rightarrow O'$

a) Calculating probability distribution

$$P(S) = \frac{3}{5} = 0.6$$

$$P(s') = \frac{2}{5} = 0.4$$

$$P(o) = \frac{4}{5} = 0.8$$

$$P(o') = \frac{1}{5} = 0.2$$

1). Joint probability distribution

$$P(s, o) = \frac{3}{5} = 0.6$$

$$P(s', o') = \frac{1}{5} = 0.2$$

$$P(s, o') = 0$$

$$P(s', o) = \frac{1}{5} = 0.2$$

c). Check if distribution are independent

Need to verify, joint probability is equal to product of their marginal probabilities.

Calc. for marginal product distribution

$$P(s) \times P(o) = \frac{3}{5} \times \frac{4}{5} = \frac{12}{25} = 0.48$$

$$P(s') \times P(o) = \frac{2}{5} \times \frac{4}{5} = \frac{8}{25} = 0.32$$

$$P(s) \times P(o') = \frac{3}{5} \times \frac{1}{5} = \frac{3}{25} = 0.12$$

$$P(s') \times P(o') = \frac{2}{5} \times \frac{1}{5} = \frac{2}{25} = 0.08$$

∴ joint probability doesn't match the product of marginals, so distributions are independent.

d) mutual information = $\sum_x \sum_y p_{x,y}(x,y) \ln \left(\frac{p_{x,y}(x,y)}{p_x(x) p_y(y)} \right)$

$p_{x,y}(x,y) \rightarrow$ joint probability distribution of having social media account & obesity.

$p_x(x) \rightarrow$ marginal probability distribution of social media.

$p_y(y) \rightarrow$ marginal probability distribution of obesity.

$$p_{x,y}(S, 0) = 3/5$$

$$p_{x,y}(S', 0) = 1/5$$

$$p_{x,y}(S, 0') = 0$$

$$p_{x,y}(S', 0') = 1/5$$

$$p(S) = 3/5$$

$$p(S') = 2/5$$

$$p(0) = 1/5$$

$$p(0') = 1/5$$

$$\begin{aligned} \text{mutual info} &= \frac{3}{5} \ln \left(\frac{3/5}{(3/5 \times 1/5)} \right) + 0 + \frac{1}{5} \ln \left(\frac{1/5}{(2/5 \times 1/5)} \right) + \frac{1}{5} \ln \left(\frac{1/5}{(2/5 \times 1/5)} \right) \\ &= \frac{3}{5} \ln(5/4) + \frac{1}{5} \ln(5/8) + \frac{1}{5} \ln(5/2) \\ &= 0.223 \end{aligned}$$

e) Theoretically, mutual info. of 2 independent random variables is 0. If X & Y are independent, then $p(x,y) = p(x) * p(y)$, which would make the logarithm term $\ln(p(x,y)/(p(x)*p(y)))$ equal to 0 & the summation over all x & y would result in mutual info. of 0.

$$\text{As, mutual info.} = \sum_x \sum_y p(x,y) \ln \left(\frac{p(x,y)}{p(x) * p(y)} \right)$$

variables are independent, so

$$p(x,y) = p(x) * p(y)$$

so substituting in above mutual info

$$\text{mutual info} = \sum_x \sum_y p(x,y) \ln(1)$$

$$\text{as } \ln(1) = 0$$

so, mutual information = 0

Question - 3

X takes values in $\{-2, -1, 0, 1, 2\}$

as, $Y = |X|$

Y takes values as $\{0, 1, 2\}$

a) $\therefore X$ has equal probability for each value

$P(X=x) = \frac{1}{5}$ for $x \in \{-2, -1, 0, 1, 2\}$

deriving Y 's probability

$$P(Y=0) = P(X=0) = \frac{1}{5}$$

$$P(Y=1) = P(X=-1) + P(X=1) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

$$P(Y=2) = P(X=-2) + P(X=2) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

Joint probability

$Y \backslash X$	-2	-1	0	1	2
0	0	0	$\frac{1}{5}$	0	0
1	0	$\frac{1}{5}$	0	$\frac{1}{5}$	0
2	$\frac{1}{5}$	0	0	0	$\frac{1}{5}$

Checking, X & Y are independent

$$P(X=x, Y=y) = P(X=x) * P(Y=y) \text{ , need to prove}$$

let $x = -2$ & $y = 2$

$$P(X=-2, Y=2) = \frac{1}{5}$$

$$P(X=-2) * P(Y=2) = \frac{1}{5} \times \frac{2}{5} = \frac{2}{25}$$

$\therefore P(X=x, Y=y) \neq P(X=x) * P(Y=y)$, X & Y are not independent random variables.

1) Correlation of X, Y

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

lets, calc. different values

$$\mu_X = E[X] = \sum x \cdot p(x) = -2 \times \frac{1}{5} - 1 \times \frac{1}{5} + 0 + 1 \times \frac{1}{5} + 2 \times \frac{1}{5} = 0$$

$$\mu_Y = E[Y] = \sum y \cdot p(y) = 0 + 1 \times \frac{2}{5} + 2 \times \frac{2}{5} = \frac{6}{5} = 1.2$$

$$E[XY] = \sum_x \sum_y xy p(X=x, Y=y)$$

$$= -2 \times 2 \times \frac{1}{5} - 1 \times 1 \times \frac{1}{5} + 0 + 1 \times 1 \times \frac{1}{5} + 2 \times 2 \times \frac{1}{5} = 0$$

$$\text{cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= 0 - 0 \times 1.2 = 0$$

$$\text{Calc, } E[X^2] = \sum_x x^2 p(X=x)$$

$$= 4 \times \frac{1}{5} + 1 \times \frac{1}{5} + 0 + 1 \times \frac{1}{5} + 4 \times \frac{1}{5} = \frac{10}{5} = 2$$

$$E[Y^2] = \sum_y y^2 p(Y=y)$$

$$= 0 + 1 \times \frac{2}{5} + 4 \times \frac{2}{5} = 2$$

$$\sigma_x^2 = E[X^2] - (E[X])^2 = 2 - 0 = 2$$

$$\sigma_x = \sqrt{2}$$

$$\sigma_y^2 = E[Y^2] - (E[Y])^2 = 2 - (1.2)^2 = \frac{14}{25}$$

$$\sigma_y = \sqrt{14/25}$$

$$\text{So, } \text{corr}(X, Y) = \frac{0}{\sqrt{2} \times \sqrt{14/25}} = 0$$

- c). The result in part (b) indicates that $\text{corr}(X, Y) = 0$, meaning there is no linear relationship b/w X & Y . Although X & Y are dependent because $Y = |X|$, this dependency is non-linear due to the absolute value function. Consequently, changes in X don't predict changes in Y in a linear manner. This demonstrates that 0 correlation reflects the absence of linear dependency, not overall dependency. The symmetric distribution of X around 0 results in 0 covariance, confirming the lack of linear correlation despite the dependence b/w X & Y .

d). The fundamental difference b/w these 2 questions (2 & 3) & the current question is the relationship b/w the random variables X & Y .

In questions 2 & 3, the random variables X & Y were independent, meaning that the knowledge of 1 variable provided no information about the other variable.

However, in this question, X & Y aren't independent random variables. Y is defined as the absolute value of X , i.e., $Y = |X|$. This functional relationship b/w X & Y introduces a dependency b/w the 2 variables.

When 2 random variables are independent, their joint probability distribution is simply the product of their marginal distributions, & their correlation (or covariance) is 0.

On the other hand, when 2 random are dependent or related, as in this case, their joint probability distribution cannot be factored into the product of their marginal distributions, & their correlation (or covariance) is generally non-zero.

This difference in the relationship b/w X & Y (independent vs dependent) leads to different calculations & interpretations of the joint probability distribution, mutual info. (as seen in ques 2), & correlation (as seen in this ques).

In summary, while ques 2 & 3 dealt with independent random variables, this ques involves dependent random variables, which necessitates a different approach & considerations in analyzing their joint behavior & statistical properties.

Question-4

a) $f(x,y) = \sqrt{2x+3y} - \frac{x}{y}$
 - compute $f(3,1)$
 $f(3,1) = \sqrt{2 \times 3 + 3 \times 1} - \frac{3}{1} = \sqrt{9} - 3 = 0$

- compute partial derivatives & then find at $(3,1)$

$$f_x(x,y) = \frac{d}{dx} (\sqrt{2x+3y} - \frac{x}{y})$$

$$= \frac{2}{2\sqrt{2x+3y}} - \frac{1}{y} = \frac{1}{\sqrt{2x+3y}} - \frac{1}{y}$$

$$f_y(x,y) = \frac{d}{dy} (\sqrt{2x+3y} - \frac{x}{y})$$

$$= \frac{3}{2\sqrt{2x+3y}} + \frac{x}{y^2}$$

Calc. at $(3,1)$

$$f_x(3,1) = \frac{1}{\sqrt{2 \times 3 + 3 \times 1}} - \frac{1}{1} = \frac{1}{\sqrt{9}} - 1 = \frac{1}{3} - 1 = -\frac{2}{3}$$

$$f_y(3,1) = \frac{3}{2\sqrt{2 \times 3 + 3 \times 1}} + \frac{3}{1^2} = \frac{3}{2\sqrt{9}} + 3 = \frac{3}{2 \times 3} + 3 = \frac{7}{2}$$

Linearized func. at $(3,1)$

$$L(x,y) = f(3,1) + f_x(3,1)(x-3) + f_y(3,1)(y-1)$$

$$= 0 - \frac{2}{3}(x-3) + \frac{7}{2}(y-1)$$

$$L(x,y) = -\frac{2x}{3} + \frac{7y}{2} - \frac{3}{2}$$

1) $f(x,y) = e^{5x} \ln(1+y)$

- compute $f(0,0)$

$$f(0,0) = e^{5 \times 0} \ln(1+0) = 1 \cdot \ln(1) = 0$$

- computing 1st order partial derivatives

$$f_x(x, y) = \frac{d}{dx} (e^{5x} \ln(1+y)) = 5e^{5x} \ln(1+y)$$

$$f_y(x, y) = \frac{d}{dy} (e^{5x} \ln(1+y)) = \frac{e^{5x}}{1+y}$$

calc. at (0,0)

$$f_x(0,0) = 5e^{5 \times 0} \ln(1+0) = 0$$

$$f_y(0,0) = \frac{e^{5 \times 0}}{1+0} = 1$$

- computing 2nd order partial derivatives

$$f_{xx}(x, y) = \frac{d^2}{dx^2} (e^{5x} \ln(1+y)) = 25e^{5x} \ln(1+y)$$

$$f_{xy}(x, y) = \frac{d^2}{dx dy} (e^{5x} \ln(1+y)) = \frac{5e^{5x}}{1+y}$$

$$f_{yy}(x, y) = \frac{d^2}{dy^2} (e^{5x} \ln(1+y)) = -\frac{e^{5x}}{(1+y)^2}$$

calc. at (0,0)

$$f_{xx}(0,0) = 25e^{5 \times 0} \ln(1+0) = 0$$

$$f_{xy}(0,0) = \frac{5e^{5 \times 0}}{1+0} = 5$$

$$f_{yy}(0,0) = -\frac{e^{5 \times 0}}{(1+0)^2} = -1$$

2nd order Taylor polynomial at (0,0) is

$$T(x, y) = f(0,0) + f_x(0,0)x + f_y(0,0)y + \frac{1}{2} \begin{pmatrix} f_{xx}(0,0)x^2 \\ + 2f_{xy}(0,0)xy \\ + f_{yy}(0,0)y^2 \end{pmatrix}$$

$$= 0 + 0 + y + \frac{1}{2} (0 + 2 \times 5xy - 1y^2)$$

$$T(x, y) = y + 5xy - \frac{y^2}{2}$$

c). $f(x, y, z) = yx^2 + zy^2 + z^2 - 2yx + 2zy + y - z$

(i) find all stationary pts. of this func.

$$f_x = \frac{d}{dx} (yx^2 + zy^2 + z^2 - 2yx + 2zy + y - z)$$

$$= 2xy - 2y = 2y(x-1)$$

$$f_y = \frac{d}{dy} (yx^2 + zy^2 + z^2 - 2yx + 2zy + y - z)$$

$$= x^2 + 2zy - 2x + 2z + 1$$

$$f_z = \frac{d}{dz} (yx^2 + zy^2 + z^2 - 2yx + 2zy + y - z)$$

$$= y^2 + 2z + 2y - 1$$

Setting partial derivatives to 0

$$2xy - 2y = 0 \quad \text{--- (1)}$$

$$x^2 + 2zy - 2x + 2z + 1 = 0 \quad \text{--- (2)}$$

$$y^2 + 2z + 2y - 1 = 0 \quad \text{--- (3)}$$

Simplifying eq. (1)

$$2y(x-1) = 0$$

$$\text{So, } y = 0, \quad x = 1$$

case 1: $y = 0$

substituting $y = 0$ in other 2 eqs.

$$x^2 - 2x + 2z + 1 = 0 \quad \text{--- (4)}$$

$$2z - 1 = 0 \Rightarrow z = \frac{1}{2}$$

substituting $z = \frac{1}{2}$ in (4)

$$x^2 - 2x + 1 + 1 = 0$$

$$x^2 - 2x + 2 = 0$$

Discriminant, $\Delta = b^2 - 4ac$

$$= (-2)^2 - 4 \times 1 \times 2 = -4$$

which is -ve, $\therefore$ no real solution for x in this case.

case 2: $x=1$
 substituting $x=1$ in other 2 eq. ② & ③
 $1+2zy-2+2z+1=0 \Rightarrow 2zy+2z=0$
 $2z(y+1)=0$
 so, $z=0, y=-1$
 $y^2+2z+2y-1=0$ — ⑤

case 2.1: $z=0$
 substituting $z=0$ into eq ⑤
 $y^2+2y-1=0$
 $y = \frac{-2 \pm \sqrt{4+4}}{2} = -1 \pm \sqrt{2}$

so, we get 2 solutions
 $(x, y, z) = (1, -1+\sqrt{2}, 0)$ OR $(1, -1-\sqrt{2}, 0)$

case 2.2: $y=-1$
 substituting $y=-1$ into ⑤
 $1+2z-2-1=0$
 $z=1$

so, we get solution as
 $(x, y, z) = (1, -1, 1)$

Stationary points are
 $(1, -1+\sqrt{2}, 0), (1, -1-\sqrt{2}, 0), (1, -1, 1)$

(ii) find Hessian matrix, H

$$H = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}$$

Computing 2nd order derivatives

$$f_{xx} = \frac{d^2 f}{dx^2} = 2y$$

$$f_{xy} = \frac{d^2 f}{dx dy} = 2x - 2 = f_{yx}$$

$$f_{xz} = \frac{d^2 f}{dx dz} = 0 = f_{zx}$$

$$f_{yy} = \frac{d^2 f}{dy^2} = 2z$$

$$f_{yz} = \frac{d^2 f}{dy dz} = 2y + 2 = f_{zy}$$

$$f_{zz} = \frac{d^2 f}{dz^2} = 2$$

$$H = \begin{bmatrix} 2y & 2x-2 & 0 \\ 2x-2 & 2z & 2y+2 \\ 0 & 2y+2 & 2 \end{bmatrix}$$

(iii) classify stationary points

- at $(1, -1 + \sqrt{2}, 0)$

$$H = \begin{bmatrix} -2+2\sqrt{2} & 0 & 0 \\ 0 & 0 & 2\sqrt{2} \\ 0 & 2\sqrt{2} & 2 \end{bmatrix}$$

$$\det(H - \lambda I) = 0$$

$$\det \begin{bmatrix} -2+2\sqrt{2}-\lambda & 0 & 0 \\ 0 & -\lambda & 2\sqrt{2} \\ 0 & 2\sqrt{2} & 2-\lambda \end{bmatrix} = 0$$

$$(-2+2\sqrt{2}-\lambda)[(-2\lambda)+\lambda^2-8] = 0$$

$$(-2+2\sqrt{2}-\lambda)(\lambda-4)(\lambda+2) = 0$$

$$\lambda = 4, \lambda = -2, \lambda = \sqrt{2} - 1$$

$\therefore$ all eigenvalues don't have same sign & all are non-zero, then the critical point is a saddle point.

- at $(1, -1-\sqrt{2}, 0)$

$$H = \begin{bmatrix} -2-2\sqrt{2} & 0 & 0 \\ 0 & 0 & -2\sqrt{2} \\ 0 & -2\sqrt{2} & 2 \end{bmatrix}$$

$$\det(H - \lambda I) = 0$$

$$\det \begin{bmatrix} -2-2\sqrt{2}-\lambda & 0 & 0 \\ 0 & -\lambda & -2\sqrt{2} \\ 0 & -2\sqrt{2} & 2-\lambda \end{bmatrix} = 0$$

$$(-2-2\sqrt{2}-\lambda)(-\lambda)^2 - 8 = 0$$

$$(-2-2\sqrt{2}-\lambda)(\lambda^2 + 4 - 4\lambda + 2\lambda - 8) = 0$$

$$(-2-2\sqrt{2}-\lambda)(\lambda-4)(\lambda+2) = 0$$

$$\lambda = 4, \lambda = -2, \lambda = -(\sqrt{2}+1)$$

$\therefore$, all eigenvalues don't have same sign & all are non-zero, then the critical point is a saddle point.

- at $(1, -1, 1)$

$$H = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\det(H - \lambda I) = 0$$

$$\begin{bmatrix} -2-\lambda & 0 & 0 \\ 0 & 2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{bmatrix} = 0$$

$$(-2-\lambda)(2-\lambda)(2-\lambda) = 0$$

$$\lambda = -2, \lambda = 2, \lambda = 2$$

$\therefore$, all eigenvalues don't have same sign & all are non-zero, then the critical point is a saddle point.

d).

$$f(x, y) = x^4 + kxy + y^4$$

- compute 1st order derivative & set to 0

$$f_x = \frac{d}{dx}(x^4 + kxy + y^4) = 4x^3 + ky$$

$$f_y = \frac{d}{dy}(x^4 + kxy + y^4) = kx + 4y^3$$

(calc. at $(1, 1)$)

$$f_x(1, 1) = 4(1)^3 + k(1) = 4 + k$$

$$f_y(1, 1) = k(1) + 4(1)^3 = k + 4$$

for $(1, 1)$ to be stationary point

$$4 + k = 0 \Rightarrow k = -4$$

So, for $(1, 1)$ to be stationary point k must be -4 .

- compute 2nd order derivative & form Hessian matrix

$$f_{xx} = \frac{d^2 f}{dx^2} = 12x^2$$

$$f_{xy} = \frac{d^2 f}{dx dy} = k$$

$$f_{yy} = \frac{d^2 f}{dy^2} = 12y^2$$

Hessian matrix

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 12x^2 & k \\ k & 12y^2 \end{bmatrix}$$

Calc. at (1,1)

$$H = \begin{bmatrix} 12 & k \\ k & 12 \end{bmatrix}$$

for (1,1) to be local min, hessian matrix must be +ve conditions for 2x2 matrix to be +ve are.

- 1) $f_{xx} > 0$
- 2) $\det(H) > 0$

So, as per above conditions

$$f_{xx} = 12 > 0$$

$$\det(H) > 0$$

$$144 - k^2 > 0$$

$$k^2 < 144 \Rightarrow |k| < 12$$

$$-12 < k < 12$$

So, for $f(x,y)$ to have local min at (1,1), k must be within $-12 < k < 12$

e). $f(x,y,z) = -x - y + z$
 subject to $x^2 + y^2 = 2$, $x + y + z = 1$

- Using Lagrange multipliers

$$\mathcal{L}(x,y,z,\lambda_1,\lambda_2) = -x - y + z + \lambda_1(x^2 + y^2 - 2) + \lambda_2(x + y + z - 1)$$

where λ_1 & λ_2 are Lagrange multipliers

- Compute partial derivatives

$$\frac{d\mathcal{L}}{dx} = -1 + 2\lambda_1 x + \lambda_2$$

$$\frac{d\mathcal{L}}{dy} = -1 + 2\lambda_1 y + \lambda_2$$

$$\frac{dL}{dz} = 1 + \lambda_2$$

setting partial derivative to 0

$$-1 + 2\lambda_1, x + \lambda_2 = 0 \quad \text{--- (1)}$$

$$-1 + 2\lambda_1, y + \lambda_2 = 0 \quad \text{--- (2)}$$

$$1 + \lambda_2 = 0 \quad \text{--- (3)}$$

from eq. (3)

$$\lambda_2 = -1$$

substituting $\lambda_2 = -1$ in (1) & (2)

$$-1 + 2\lambda_1, x - 1 = 0 \Rightarrow x = \frac{1}{2}\lambda_1$$

$$-1 + 2\lambda_1, y - 1 = 0 \Rightarrow y = \frac{1}{2}\lambda_1$$

using, $x^2 + y^2 = 2$

$$\left(\frac{1}{2}\lambda_1\right)^2 + \left(\frac{1}{2}\lambda_1\right)^2 = 2$$

$$\lambda_1 = \pm 1$$

$$\text{if } \lambda_1 = 1, x = y = 1$$

$$\text{if } \lambda_1 = -1, x = y = -1$$

Now using $x + y + z = 1$

- for $x = y = 1$

$$1 + 1 + z = 1$$

$$z = -1$$

so, 1 point is $(1, 1, -1)$

- for $x = y = -1$

$$-1 - 1 + z = 1$$

$$z = 3$$

so, another point is $(-1, -1, 3)$

- evaluate $f(x, y, z)$ at critical points

$$f(1, 1, -1) = -1 - 1 - 1 = -3$$

$$f(-1, -1, 3) = -(-1) - (-1) + 3 = 5$$

min. value of $f(x, y, z)$ is -3 at $(1, 1, -1)$

max. value of $f(x, y, z)$ is 5 at $(-1, -1, 3)$